Credit Application Process

Step-by-step textual description of the BPMN workflow

This document describes the sequence modeled in Credit Application.bpmn and the associated Credit application, Project analysis, and Credit File Analysis forms. It follows the modeled business workflow from the initial application to the final customer notification.

Part 1 - Verification of the customer's file

The process begins at the “Credit Application” start event. A parallel gateway immediately starts two branches that run at the same time. First, the customer completes the credit application form. In the second, the bank records the request, collects the supporting documents, and carries out two automated checks: completeness of the application, then customer eligibility. A file that fails either check is rejected, and the customer is notified. A file that passes both checks waits at a second parallel gateway until the customer’s form is also complete. Only then does project evaluation begin. In the supplied implementation, both checks are simulated with random results rather than real verifications.

Step 1: Credit Application start event

The process begins at the “Credit Application” start event. A parallel gateway immediately starts two activities at the same time.

Step 2: Collect the customer’s application details

The customer completes the “fulfill of the form by the client” user task using the Credit application form. The form requires the requested loan amount, monthly income, total savings, desired loan term in months, a description of the financing project, and employment status (employee or unemployed). All six inputs are mandatory. The minimum permitted loan term is 60 months, or five years. Submitting the form makes these values available to later assessment tasks.

Step 3: Register the application and receive documents

In the parallel administrative branch, the bank first performs “Record the request and open the case”, then “Receive the supporting documents”. These tasks represent the registration of the financing request and collection of the evidence necessary for review. The BPMN does not enumerate a specific list of required supporting documents.

Step 4: Check application completeness

The “Verify the completeness of the application” script task produces a compliance result. At the gateway “Is the application complete and in order?”, a result of CONFORME sends the file onward; NON CONFORME sends it to “Notify the customer of the rejection” and then the “Rejection of the application” end event. In the supplied implementation, the script currently generates this result randomly rather than actually checking documents.

Step 5: Archive documents and verify customer eligibility

For a compliant application, the bank performs “Archive the documents” and then “Verify the customer’s identity, age, and possible banking ban”. The eligibility gateway examines the result: ELIGIBLE allows the file to continue; NON ELIGIBLE causes a customer rejection notification and ends that administrative branch. As modeled, the eligibility result is also generated by a random script rather than by live identity or banking checks.

Step 6: Synchronize both initial branches Lucas

A parallel gateway joins the completed customer form branch and the successful administrative verification branch. Project evaluation begins once the customer has supplied the requested information and the application has passed the completeness and eligibility checks.

Part 2 - Project Evaluation

The project evaluation phase begins after the customer's application has been completed and the administrative checks have confirmed that the file is compliant and the customer is eligible. A banking advisor first assesses the financing project and records an approval decision, a project grade, and a written justification. If the project is approved, automated tasks calculate the financial indicators, check for obstructing criteria, and combine the financial consistency score with the project grade. The resulting overall score determines whether the credit is automatically approved, automatically rejected, or referred to an advisor for a final manual review. In every case, the customer is informed of the decision.

Step 7: Review the proposed financing project

The "Verify the project" user task opens the Project analysis form for a banking advisor. The advisor reads the customer's amount requested, income, savings, loan term, project description and employment status. They must select Project Approved or Project Rejected, give the project a grade from 1 to 10, and write a justification (stored in justificationConseiller). If the project is rejected (REFUSE), the file goes to the credit-rejection notification and end event. If it is approved (VALIDE), the financial calculations begin.

Step 8: Calculate the financial indicators

Three automated script tasks execute sequentially for an approved project. First, "Calculate Loan Coverage by Savings" measures savings relative to the amount requested. Second, "Calculate Mensuality" divides the requested amount by the loan duration to estimate a monthly payment. Third, "Calculate the debt-to-income ratio" compares that estimated monthly payment with monthly income.

Indicator	Formula in the BPMN
Savings coverage (%)	(Savings ÷ Amount requested) × 100
Monthly payment (€)	Amount requested ÷ Loan term (months)
Debt-to-income ratio (%)	(Monthly payment ÷ Monthly income) × 100

These calculations are simplified: the modeled monthly payment does not include interest, insurance or fees, and the debt ratio does not incorporate existing monthly debt obligations.

Step 9: Check for an obstructing financial criterion

The "Obstructing criterion?" gateway checks the debt-to-income ratio and the savings coverage. The YES branch leads to credit rejection when at least one obstructing condition is met: the debt-to-income ratio is above 50% OR the savings coverage is below 10%. The NO branch allows the application to proceed to scoring only when both favorable conditions are met: the debt-to-income ratio is at or below 50% AND the savings coverage is at or above 10%. These complementary conditions also cover mixed outcomes. For example, a debt ratio of 40% with savings coverage of 5% takes the YES (rejection) branch, while a debt ratio of 40% with savings coverage of 20% takes the NO (continue scoring) branch.

Step 10: Calculate the financial consistency score

The “Calculate the consistency score” script assigns points according to debt-to-income ratio and savings coverage. It then adds a bonus if the debt-to-income ratio is especially low. The resulting scoreCoherence has a maximum of 100 points.

Debt-to-income ratio	Points	Savings coverage	Points
≤ 30%	50	≥ 75%	40
> 30% to 35%	40	≥ 50% to < 75%	30
> 35% to 40%	25	≥ 25% to < 50%	20
> 40% to 50%	10	≥ 10% to < 25%	10
> 50%	0	< 10%	0

An additional 10 points are granted when the debt-to-income ratio is at or below 10%. Thus: Consistency score = debt-ratio points + savings-coverage points + low-debt bonus.

Step 11: Calculate the overall score

The “Calculate the overall score” script combines the automated consistency score with the advisor’s project grade. The consistency score receives 70% of the weighting and the project grade receives 30%; the grade is multiplied by 10 to express it on a 100-point scale.

Overall score = (Consistency score × 0.70) + (Project grade × 10 × 0.30).

For example, a consistency score of 80 and a project grade of 7 produce an overall score of 77: (80 × 0.70) + (7 × 10 × 0.30) = 77.

Step 12: Apply the automatic decision rules

At the “Automatic Decision” gateway, an overall score below 50 results in automatic rejection; a score of at least 70 results in automatic approval; and a score from 50 inclusive to below 70 is sent for additional manual analysis. The first two paths proceed directly to their corresponding customer notification tasks.

Overall score	Modeled outcome
Below 50	Automatic rejection
50 to below 70	Manual credit-file review
70 or higher	Automatic approval

Step 13: Manually review intermediate-score applications

When the score is between 50 and 70, the “Manually analyze the file” user task opens the Credit File Analysis form. The advisor can review the consistency score, project grade, debt-to-income ratio, overall score, requested amount, monthly income and monthly payment. They must choose “Grant the credit” (ACCORD) or “Reject the credit” (REFUS) and provide a mandatory written justification. The “Banking Advisor Decision” gateway then routes the file to approval or rejection according to that selection.

Step 14: Notify the customer and finish the process

On the approval path, the bank executes “Notify the client of the agreement” before reaching the “Credit granted” end event. On the project-evaluation or credit-scoring rejection path, it executes “Notify the client of the rejection” before reaching the “Credit rejected” end event. Rejections during the earlier administrative checks use their own rejection notification and “Rejection of the application” end event. The modeled process ends with the decision and its communication; it does not cover contract signing or fund disbursement.